MILITARY SPECIFICATION
COUNTER, PULSE, ELECTRONIC, 18 GIGAHERTZ (GHz)
TD-1338( )/USM

This specification is approved for use by the US Army Communications Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers solid state automatic 18 GHz Pulsed Counters hereinafter called equipment.

1.2 Classification. Equipment covered by this specification may be commercially available equipment modified to the extent necessary to meet the following description. The equipment will be as defined in MIL-T-28800 and herein for Type III, Class 5, Style E and color optional with manufacturer.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids, or the request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-28777	Cable Assembly, Electronic Test Equipment (3 Wire, 3 Prong, 125 Volt, Molded) Grounding Plug Connectors
MIL-T-28800	Test Equipment For Use With Electrical And Electronic Equipment, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Communications Research and Development Command, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-471	Maintainability/Verification/Demonstration/ Evaluation
MIL-STD-781	Reliability Tests, Exponential Distribution

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS

IEEE-STD-488-1978	Standard Digital Interface for Programmable Instrumentation

(Application for copies should be addressed to the IEEE Service Center, 445 Hoes Lane, Piscataway, New Jersey 08854)

3. REQUIREMENTS

3.1 Classification of requirements. The requirements for the equipment are classified as follows:

REQUIREMENT	PARAGRAPH
Safety	3.2
Parts, materials, and processes	3.3
Design and construction	3.4
Electrical power sources and connections	3.5
Enclosure requirements	3.6
Marking and identification	3.7
Environmental requirements	3.8
Reliability	3.9
Maintainability requirements	3.10
Performance requirements	3.11

3.2 Safety. Unless otherwise specified herein, the equipment shall comply with the Type III safety requirements specified in MIL-T-28800.

3.3 Parts, materials and processes Unless otherwise specified herein the equipment shall comply with Type III parts, materials and processes requirements of MIL-T-28800.

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3.3.1 Restricted materials. Equipment shall comply with the restricted material requirements of MIL-T-28800 including the requirement for mercury or radioactive materials and shall contain no combination of materials which cause deterioration of any material contained in the equipment due to effects of outgassing.

3.3.2 Input and output signal connectors. Except for the logic interface and the microwave frequency input, all input and output connectors shall be series BNC female. The logic interface connector shall be in accordance with 3.11.10. A Type N Connector shall be used for the microwave frequency input.

3.4 Design and construction. Unless otherwise specified herein, the equipment shall comply with the Type III, Class 5, Style E, Color R, design and construction requirements of MIL-T-28800 with the convertible/rack-mountable capability.

3.4.1 First article submission. When required in the contract order or solicitation, sample unit(s) shall be furnished for First Article inspection and approval.

3.4.2 Mainframe plug-in concept. Equipment utilizing externally accessible, externally removable subassemblies (mainframe plug-in concept) is acceptable only if the plug-in(s) are not removable by access through the front panel or rear panel with the unaided hand. A system part number shall be assigned to any such acceptable mainframe plug-in combination. The system part number shall be marked on the mainframe identification plate and any plug-in(s).

3.4.3 Controls. Unless otherwise specified herein, built-in adjustments and compensating devices shall not be externally accessible. All potentiometer controls shall have at least a 4.7 radian (270 degrees) rotational range.

3.4.3.1 Front panel controls. Controls for the following functions shall be located on the front panel.

a. Sample rate

b. Resolution

c. Reset

d. Display segment test

e. Self check

f. Input port selection

3.4.4 Accessibility. The equipment shall be constructed so that:

a. Subassemblies and chassis components can be removed without removing any other hard wired printed circuit cards, subassembly, or component.

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b. Adjustments can be made without removing any component, printed circuit card or subassembly except the use of extender cards is permitted.

c. Printed circuit cards can be removed for servicing or adjustments without the need to unsolder cables and interconnecting wiring. Printed circuit cards (mother boards) designed primarily to distribute power and signals to other printed cards (daughter boards) are excluded from this requirement. When such mother boards are used, they shall be accessible from both sides.

3.4.5 Encapsulation and embedment. Encapsulation and embedment (potting) of subassemblies shall not be used.

3.5 Electrical power sources and connections.

3.5.1 Electrical Power Sources. The equipment shall comply with the Type III 50 to 60 Hertz (Hz) single phase 115/230 Volt (V) and 400 Hz single phase 115V alternate power sources requirements of MIL-T-28800.

3.5.2 Maximum power. The equipment shall consume not more than 200 watts.

3.5.3 Power connections. The equipment shall comply with the Type III power connections requirements of MIL-T-28800 except as specified herein.

3.5.3.1 Power Cables.

a. For 115 VAC operation, the power cables shall be in accordance with MIL-C-28777/3 except that the molded connector at the equipment end shall mate with Switchcraft Part Number EAC 302 connector or equal.

b. For 230 VAC operation, the power cables shall be as specified in the contract or order. (See 6.2.1.2)

3.5.4 Input power selection device. An input power selection device shall be provided for selection of input power voltages of 115 VAC or 230 VAC. Provision shall be incorporated to prevent accidental switching. The power selection device shall be mounted on the rear panel. When the equipment is delivered, the power selection device shall be in the 115 VAC position.

3.5.5 Fuses and circuit breakers. Fuses or circuit breakers shall be in accordance with MIL-T-28800. Either common or separate fuseholders may be provided. If only one fuseholder is used (common), the equipment shall be provided with the 115 VAC fuse installed and the 230 VAC fuse shall be stowed with the accessories.

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3.5.6 Power indicator. A front panel light emitting indicator shall be provided to indicate when the equipment is energized.

3.6 Enclosures. The equipment enclosure shall be as specified in MIL-T-28800 for Style F.

3.7 Marking and identification. The equipment shall comply with the Type III marking and identification requirements of MIL-T-28800.

3.8 Environmental requirements. The equipment shall comply with the Class 5 environmental requirements of MIL-T-28800 except as required herein.

3.8.1 Relative Humidity. The humidity requirement is extended up to 95 percent of relative humidity as required for Class 4 equipments throughout the entire operating range of the equipment.

3.9 Reliability. The minimum acceptable value (MAV) of mean-time-between-failure (MTBF) shall be 2000 hours.

3.9.1 Burn-In. Each equipment delivered against this specification shall be submitted to a minimum 96 hour on-time burn-in procedure as specified in 4.4.6.1. The last 24 hours of burn-in shall be failure free.

3.10 Maintainability requirements. The equipment shall comply with the Type III maintainability requirements of MIL-T-28800 and as specified herein (See 6.2.3.2). The mean corrective maintenance time (M0 in MIL-STD-471) shall be 90 minutes. The maximum tolerable mean corrective maintenance time (M1 in MIL-STD-471) shall be 15 minutes. The maximum corrective maintenance time shall not exceed 270 minutes. The mean and maximum corrective maintenance times shall include all time required to troubleshoot, fault isolate, repair, and to calibrate the equipment for any malfunction down to the lowest discrete component (resistor, switch, transistor, integrated circuit, non-repairable assembly, and so forth), of the equipment, but does not include certification time (See 3.10.1). Unless otherwise specified in contract, maintainability demonstration during quality conformance inspection shall not be required.

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3.10.1 Certification time. Certification time shall not exceed 2.0 hours, and it shall include all time required to check all inputs and outputs, on all range function settings.

3.10.2 Maintenance provisions. The equipment shall meet the maintenance provisions of MIL-T-28800 except the accessibility of accessories does not apply.

3.10.2.1 Maintenance and calibration aids. When required in the contract or order, any circuit board extenders, special adapters, special tools, and patch cables required for maintenance or calibration of equipment shall be stowed in suitable reusable pouch or container which shall be marked "MAINTENANCE AIDS FOR USE WITH (equipment nomenclature or model number)" (See 6.2.1.4). The equipment nomenclature shall be used thereon if assigned (See 6.2.2.1).

3.11 Performance requirements. The equipment shall contain the following electrical performance capabilities.

a.	Warm-up performance	(3.11.1)
b.	Time base	(3.11.2)
c.	Display	(3.11.3)
d.	Sample rate	(3.11.4)
e.	Self check	(3.11.5)
f.	Frequency measurement-pulsed	(3.11.6)
g.	Frequency measurement - CW	(3.11.7)
h.	Input characteristics	(3.11.8)
i.	Outputs	(3.11.9)
j.	Logic interface	(3.11.10)
k.	Accessories	(3.11.11)

3.11.1 Warm-up performance. Warm-up time shall not exceed one minute.

3.11.2 Time base. Time base specifications are given in terms of fractional frequency error (ffe) (See 6.3.1). Except for temperature stability, all time base requirements are specified for 25°C ± 0.5°C. Time base requirements are specified for measurement from the 10 megahertz (MHz) standard output with the equipment in the self test mode (See 4.5).

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3.11.2.1 Time base stability. Unless otherwise required in the contract or order, the equipment shall comply with at least the following requirements.

3.11.2.1.1 Long term stability (aging rate). The aging rate shall be less than 1 X 10-8 per 24 hours after 48 hours of operation (M8, M9) preceded by at least 72 hours with no source power applied.

3.11.2.1.2 Frequency retrace stability. The ffe 30 minutes after application of source power (M2), following at least a 72 hour period of nonoperation, shall be within ±2 X 10-6 of the ffe immediately prior to the 72 hour period of nonoperation (M1). The ffe 5 hours after application of source power (M4), following at least a 72 hour period of nonoperation, shall be within ±1 X 10-6 of the ffe immediately prior to the 72 hour period of nonoperation (M1). The 72 hour period of nonoperation shall be preceded by a least 72 hours of operation.

3.11.2.1.3 Restabilization, stability. The ffe 5 hours after application of source power (M6) shall be within ±5 X 10-6 of the ffe measured 48 hours after application of source power (M7). Application of source power shall be preceded by at least 72 hours of nonoperation.

3.11.2.1.4 Short term stability. The short term stability shall be less than ±2 X 10-8 root mean square (rms) 30 minutes after application of source power (M3) and less than ±2 X 10-9 rms 5 hours after application of source power (M5). Application of source power shall be preceded by a least 72 hours with no source power applied.

3.11.2.1.5 Stability during power change. The ffe shall not change more than ±1 X 10-6 during the first two minutes after any power change within the specified steady state band nor more than 1 X 10-7 at the end of the two minutes (See 3.5.1).

3.11.2.1.6 Stability during temperature change. The ffe shall not change more than ±2 X 10-6 total for any incremental or total temperature change within the specified temperature operating range (See 3.8.).

3.11.2.1.7 Total ffe stability. The total calculated ffe at the end of a 12 month period shall not exceed ±6 X 10-6.

3.11.2.1.8 Oscillator adjustments stability. The oscillator shall be capable of being adjusted so that the 10 MHz standard output shall be within ±1 part in 106 of an external 10 MHz standard. The total range of adjustment shall be at least ±1 part in 106 from 10 MHz.

3.11.2.2 External standard input. Equipment shall be capable of specified operation from any 10 MHz external frequency standard having a value of 1 volt rms into 50 ohms. When the external standard is used, it shall be in lieu of the internal standard.

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3.11.2.2.1 External standard input impedance The load impedance at the equipment external standard input port shall be not less than 1 kilohm paralleled by not more than 30 picofarad (pf).

3.11.3 Display. All front panel display devices shall be readable at a distance of 15 feet through a subtended angle of 45 degrees. Provisions shall be incorporated to test all light emitting segments of the display devices.

3.11.3.1 Numeric readout. The numeric readout shall be at least 7 character, in line, horizontal, single plane, and digital display.

3.11.3.1.1 Measurement units. Measurement units, GHz, MHz and KHz shall be presented in conjunction with the numeric display.

3.11.3.1.2 Display storage. The numeric readout shall be continuous and change only when the measurement is different from that counted during the previous measurement cycle.

3.11.3.1.3 Display blanking All non-significant leading zeros shall be blanked. All readout digits below the selected resolution of the displayed frequency shall not be displayed.

3.11.3.2 Gate lamp. A front panel display device shall indicate gate opening and closing.

3.11.4 Sample rate The sample rate shall be variable from 0.1 to at least 5 seconds. It shall be independent of resolution control and capable of maintaining the last reading or the display until the "reset" button is pressed Erroneous data resulting from conflicts between sample rate and gate time shall not be displayed (See 6.3.3).

3.11.4.1 Reset. The reset control shall allow manual initiation of the gate

3.11.5 Self check. The self check function shall be provided to verify functional operation of the frequency measurement.

3.11.6 Frequency Measurement - pulse modulated signals.

3.11.6.1 Frequency range. The frequency range for automatic measurement of pulse modulated signals shall be at least 400 MHz to 18 GHz.

3.11.6.2 Frequency accuracy. The frequency measurement accuracy for measurement of pulse modulated signals shall be ± 2 x LSD ± rms jitter ± gate error ± timebase error. The rms jitter shall be equal to or less than

$$\frac{2}{\sqrt{(\text{GATE TIME})\ (\text{GATE WIDTH})}} + 100 \text{ Hz rms}$$

The worst case gate error shall be $\frac{40 \text{ KHz}}{\text{GATE WIDTH}}$.

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3.11.6.3 Frequency resolution. Resolution for pulsed frequency measurements shall be defined as the rms deviation between successive measurements on a constant input frequency. The resolution shall be ± 2 x LSD ± rms jitter or better.

3.11.6.4 Minimum pulse width. The minimum rf burst width which can be automatically acquired shall be at least 100 nanoseconds defined at the points where the input burst level exceeds the counter sensitivity.

3.11.6.5 Minimum pulse repetition frequency (PRF). The minimum PRF of signals which can be automatically acquired shall be 50 Hz or lower.

3.11.6.6 Maximum pulse repetition frequency (PRF). The maximum PRF of signals which can be automatically acquired shall be 2 MHz or greater.

3.11.6.7 Measurement time. Total measurement time will depend upon burst width, pulse repetition frequency, and GATE TIME. For signals with a PRF greater than 100 Hz the total measurement time shall be less than

$$\frac{100\ \mu s + 2 \times \text{GATE TIME}}{\text{GATE WIDTH} \times \text{PRF}} + 1\ \text{second}$$

3.11.7 Frequency measurement - CW signals.

3.11.7.1 Frequency range. The frequency range shall be at least 400 MHz to 18 GHz for automatic measurement of CW signals.

3.11.7.2 Frequency accuracy. The frequency measurement accuracy shall be ± one count ± time base stability.

3.11.7.3 Frequency resolution. Resolution from at least 10 KHz to 1 MHz shall be provided.

3.11.7.4 F. M. Tolerance. The equipment shall tolerate and meet the accuracy requirements of 3.11.7.2 for signals between 400 MHz and 18 GHz which have peak to peak deviation of up to 40 MHz at a deviation rate from zero to at least 10 MHz.

3.11.7.5 Measurement time. At 10 KHz resolution, the equipment shall acquire and measure the input signal in not more than 1.4 seconds.

3.11.8 Input characteristics. A maximum of three input ports shall be provided with a switching device for operator selection of the desired input port.

3.11.8.1 Input impedance. The low impedance input port(s) shall have a nominal input impedance of 50 ohms.

3.11.8.2 Standing wave ratio SWR. The SWR of the low impedance input ports shall not exceed 2.5:1 up to 10.0 GHz and 3.5:1 from 10.0 GHz to 18 GHz.

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3.11.8.3 Coupling. AC coupling shall be provided on the BNC input port. For the Type N input, the coupling will be DC.

3.11.8.4 Sensitivity. The equipment sensitivity for the input port(s) shall be -10 dBm or less from 400 MHz to 10 GHz and -5 dBm or less from 10 GHz to 18 GHz.

3.11.8.5 Maximum operating input. The equipment shall operate satisfactorily with input signals up to +5 dBm on the input port(s)

3.11.8.6 Maximum input without damage. The equipment shall accept without damage input signals of up to +25 dBm on the input port(s).

3.11.9 Outputs.

3.11.9.1 Standard frequency outputs. The equipment shall provide a 10 MHz standard frequency output from the controlling standard which is independent of the time base control setting and is uninterrupted during all functions. The output voltage shall be at least 1 volt RMS into an external impedance of 50 ohms.

3.11.9.2 Gate output. The equipment shall provide a gate output during the frequency measurement.

3.11.9.3 Detected IF output. The equipment shall provide a detected IF output representing the envelope of the input RF burst. It should be possible to measure Pulse Repetition Frequency (PRF), Pulse Repetition Interval (PRI) and Pulse Width (PW) of the input pulse using this output and a universal counter capable of measuring period and time interval.

3.11.10 Logic interface. A logic interface shall be provided in accordance with IEEE-STD-488-1978

3.11.10.1 Function programming characteristics. All measurement data, including multipliers and units, shall be transferred by means of the logic interface. All control and switch functions except for the following shall be programmable.

a. Source power selection

b. Frequency standard selection.

c. Standby/operate selection.

d. Input impedance selection.

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3.11.10.1.1. Sample programming. Under interface control the equipment shall have two sampling modes (as selected by the controller). In the first mode the equipment shall not make any measurement until commanded to do so by the controller. The rate at which the controller can command the equipment to make measurements shall be at least 80 measurements per second or the maximum rate possible with the required resolution.

In the second mode the equipment shall continually make measurements on its own at a rate of at least 100 measurements per second or at the maximum rate possible with the required resolution.

3.11.10.2 Interface characteristics. The equipment shall include the interface functions as in a thru f.

a. Source handshake interface function (SH).

b. Acceptor handshake interface function (AH).

c. Talker interface function (T).

d. Listener interface function (L).

e Service request interface function (SR).

f. Remote local interface function (RL).

3.11.10.2 1 Numeric data. Unless otherwise specified in the contract or order, all numeric data (input and output) shall be American Standard Code for Information Interchange (ASCII) and the most significant digit shall be transferred first.

3.11.10.2.2 Alpha-numeric data Unless otherwise specified in the contract or order, all alpha-numeric data (input and output) shall be ASCII.

3.11.10.2.3 Address assignment. Complete five bit switchable address capabilities shall be provided for the talker function (T) address and the listener function (L) address on the rear panel.

3.11.11 Accessories The following accessories shall be supplied with each equipment.

a. Power cable in accordance with 3.5.3.1.

b. One operator's manual.

c. Fuse in accordance with 3.5.5.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions. Quality assurance terms used herein shall be as in MIL-T-28800.

4.2 Classification of inspections. The inspections specified herein are classified as in (a) and (b).

(a) First Article inspection (See 3.4.1 and 6.2.1).

(b) Quality conformance inspection (See 4.4).

4.2.1 First Article inspection groups. First Article examinations and tests are classified into groups in accordance with Table 1,4.3, and (a) through (f) (See 6.2.1.6).

(a) First Article inspection, Group A (See 4.3.1.1).

(b) First Article tests, Group B (See 4.3.1.2).

(c) First Article tests, Group C (See 4.3.1.3).

(d) First Article tests, Group D (See 4.3.1.4).

(e) First Article tests, Group E (See 4.3.1.5).

(f) First Article tests, Group F (See 4.3.1.6).

4.2.2 Quality conformance inspection groups. Quality conformance examinations and tests are classified into groups in accordance with Table 1, 4.4, and (a) through (f).

(a) Production inspection, Group A (See 4.4.1).

(b) Production control tests, Group B (See 4.4.2).

(c) Environmental and electrical power tests, Group C (See 4.4.3).

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(d) Environmental tests, Group D (See 4.4.4)

(e) Other tests, Group E (See 4.4.5).

(f) Reliability tests, Group F (See 4.4.6.2).

4.3 First Article inspection. Unless otherwise required herein, the test methods, procedures and conditions used for First Article inspection shall be performed on the units as specified in 4.5 and MIL-T-28800 (See 6.2.2.3, and 6.2.2.4). When the contract calls for First Article Testing to be performed by the Government, the First Article Requirements of MIL-T-28800 shall be waived.

4.3.1 First Article inspection. When required in the solicitation, the First Article inspection shall be performed on three equipments and on a sample of the front panel (See 6.3.4).

4.3.1.1 First Article inspection Group A. The three equipments shall be subjected to the inspection specified in Table I and II.

4.3.1.2 First Article tests Group B. The three equipments that have satisfactorily passed the required tests in accordance with 4.3.1.1 shall be subjected to the Group B tests of Tables I and III.

4.3.1.3 First Article tests, Group C. Three equipments that have satisfactorily passed the Group B tests shall be subjected to the Group C tests of Table I.

4.3.1.4 First Article tests, Group D. Two equipment that have satisfactorily passed the Group E tests shall be subjected to the Group D tests of Table I.

4.3.1.5 First Article tests, Group E. Equipments that have satisfactorily passed the Group B tests shall be subjected to the Group E tests of Table I. The number of equipment tested shall be as in (a) through (d).

(a) Maintainability. One equipment shall be tested.

(b) Acoustic noise. One equipment shall be tested, except that equipment that does not contain a motor (such a fan or blower) need not be tested.

(c) Mechanical stability. One equipment shall be tested.

(d) Front panel marking. One 100mm (4 inch) by 100mm (4 inch) sample, which is representative of the equipment front panel and marking, shall be tested.

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TABLE I Examination and test groups

Description	Requirement	Inspection Method	
		First Article	Quality Conformance
Group A			
Preoperational inspection	3	4.5	4.5
Level A performance	3.11	4.5	4.5
Leakage current	3.2	4.5	4.5
Group B			
Level B performance	3.11	4.5	4.5
Group C			
Temperature and humidity			
Humidity	3.8	4.5	4.5
Low temperature	3.8	4.5	4.5
High temperature	3.8	4.5	4.5
Temperature stability, Time base	3.11.2.1.6	4.5	4.5
Altitude	3.8	4.5	4.5
Vibration	3.8	4.5	4.5
Shock			
Shock pulse	3.8	4.5	4.5
Transit drop	3.8	4.5	4.5
Bench handling	3.8	4.5	4.5
Water resistance			
Dripproof	3.8	4.5	4.5
Electrical power			
Input power consumption	3.5.2	4.5	4.5
Voltage and frequency variation	3.5.1	4.5	4.5
Voltage transient	3.5.1	4.5	4.5
SWR	3.11.8.2	4.5	4.5
Line voltage stability, Time Base	3.11.2.1.5	4.5	4.5
Verification, Group C	3.11	4.5	4.5

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TABLE I Examination and test groups - Continued

Description	Requirement	Inspection Method	
		First Article	Quality Conformance
Group D			
EMI	3.8.	4.5	4.5
Group E			
Maintaintainability	3.10	4.5	4.5
Acoustic noise	3.8	4.5	4.5
Front panel marking	3.7	4.5	4.5
Group F			
Reliability	3.9	4.5	4.5

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TABLE II Level A Performance Parameters *

Parameter	Requirement
Display	3.11.3
Sample Rate	3.11.4
Self Check	3.11.5
Frequency Measurement - Pulsed	3.11.6
Frequency Measurement - CW	3.11.7
Input Characteristics	3.11.8
Outputs	3.11.9
Logic Interface	3.11.10

* Reduced Testing

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TABLE III Level B Performance Parameters

Parameter	Requirement
Warm-up	3.11.1
Time base	3.11.2
Time base, stability	3.11.2.1
Long term stability	3.11.2.1.1
Frequency retrace	3.11.2.1.2
Restabilization	3.11.2.1.3
Short term stability	3.11.2.1.4
Stability during power change	3.11.2.1.5
Stability during temperature change	3.11.2.1.6
Total ffe	3.11.2.1.7
Oscillator adjustment	3.11.2.1.8
External standard input	3.11.2.2
External standard input impedance	3.11.2.2.1
Display	3.11.3
Numeric readout	3.11.3.1
Measurement units	3.11.3.1.1
Display storage	3.11.3.1.2
Display blanking	3.11.3.1.3
Gate lamp	3.11.3.2
Sample rate	3.11.4
Reset	3.11.4.1
Self check	3.11.5
Frequency measurement - Pulse modulated signals	3.11.6
Frequency range	3.11.6.1
Frequency accuracy	3.11.6.2
Frequency resolution	3.11.6.3
Minimum pulse width	3.11.6.4
Minimum pulse repetition frequency	3.11.6.5
Maximum pulse repetition frequency	3.11.6.6
Measurement time	3.11.6.7

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TABLE III Level B Performance Parameters - Continued

Parameter	Requirement
Frequency measurement - CW	3.11.7
Frequency range	3.11.7.1
Frequency Accuracy	3.11.7.2
Frequency Resolution	3.11.7.3
FM tolerance	3.11.7.4
Measurement time	3.11.7.5
Input characteristics	3.11.8
Input impedance	3.11.8.1
Coupling	3.11.8.3
Sensitivity	3.11.8.4
Maximum operating input	3.11.8.5
Maximum input without damage	3.11.8.6
Outputs	3.11.9
Standards output	3.11.9.1
Gate output	3.11.9.2
Detected IF output	3.11.9.3
Logic interface	3.11.10
Function programming characteristics	3.11.10.1
Sample programming	3.11.10.1.1
Interface characteristics	3.11.10.2
Numeric data	3.11.10.2.1
Alpha - numeric data	3.11.10.2.2
Address assignment	3.11.10.2.3

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4.3.1.6 First Article tests, Group F. The Group F reliability demonstration test shall be conducted on three equipments in accordance with the Type III demonstration phase of MIL-T-28800.

4.4 Quality conformance inspection. Unless otherwise required herein the test methods, procedures, and conditions used for quality conformance inspection shall be as specified in 4.5 and MIL-T-28800. (See 6.2.2.9 and 6.2.2.10)

4.4.1 Production inspection. Each equipment shall be subjected to the Group A inspection of Tables I and II to ensure qualitatively the proper functioning of the equipment, including all operating controls, and conformance with safety requirements.

4.4.2 Production control tests. The Group B tests of Tables I and III shall be performed on equipment that have passed Group A inspection (See 4.4.1).

4.4.2.1 Production control sampling plan. From each lot a random sample of equipment shall be selected and subjected to the Group B tests. Minimum sampling shall be in accordance with MIL-T-28800.

4.4.3 Environmental and electrical power tests. The Group C tests of Table I shall be performed on First Production Lot equipments only which have passed the required tests in accordance with 4.4.2.

4.4.3.1 Environmental and electrical power sampling plan. From the First Production Lot only a random sample of equipment shall be selected and subjected to the Group C tests. The sampling shall be in accordance with MIL-T-28800.

4.4.4 Environmental tests. The Group D tests of Table I shall be performed on equipment which have passed the required tests in accordance with 4.4.2.

4.4.4.1 Environmental tests sampling plan. From the First Production Lot only a random sample of equipment shall be selected and subjected to the Group D tests. The sampling shall be in accordance with MIL-T-28800

4.4.5 Other tests. The Group E tests of Table I shall be performed on equipment which have passed the required tests in accordance with 4.4.2

4.4.5.1 Other tests sampling plan. From the First Production Lot only, a random sample shall be selected and subjected to the Group E tests. The sampling shall be in accordance with (a) through (c).

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(a) Maintainability. No maintainability demonstration is required

(b) Acoustic noise. One equipment.

(c) Mechanical stability. One equipment.

4.4.6 Reliability/Burn-In. Reliability & Testing and Burn-In procedures shall be performed as follows:

4.4.6.1 Burn-In Each deliverable equipment shall be subjected to a minimum 96 hours on time burn-in period, prior to Group A testing. During the last 24 hours of burn-in, the equipment must operate failure free. Up until this time, equipment will be allowed to accumulate failures. Each equipment which fails during the final 24 hour period shall be repaired and returned to test until it successfully survives a 24 hour period with out failure. Failures which occur during the burn-in test shall be noted and reported, but shall not count toward the establishment of equipment MTBF. Prior to burn-in, the satisfactory operation test of 4.5b shall be conducted. Daily satisfactory operation checks of 4.5a shall be conducted. For the last 24 hour failure free period, a complete satisfactory operation test shall be conducted prior to and after the period. While under burn-in the equipment shall be subjected to the conditions specified by Test Level B, MIL-STD-781. For purposes of burn-in only, the vibration requirement of Test Level B may be waived if each equipment is subjected prior to burn-in to the following vibration. The equipment shall be firmly secured to an L.A.B. package tester. The package tester speed shall be adjusted to 300 RPM. The equipment shall be vibrated for two hours at a one inch double amplitude.

4.4.6.2 Reliability testing. The Group F reliability tests shall be performed on the first production lot only. From those units of the first lot that have passed Group A and B tests, a random sample of ten (10) units shall be selected for Group F testing. If first article reliability demonstration was successfully run the reliability test may be waived at the discretion of the Government. It may also be waived at the discretion of the Government upon the acceptance by the Government of the Reliability Rationale (See 6.4).

4.4.6.2.1 Reliability test shall be conducted in accordance with MIL-STD-781 Test Level B, Test Plan IV.

4.4.6.2.2 Reliability examination and test method. The following inspections shall be used to verify equipment operation during reliability testing. The satisfactory operation test of 4.5.b shall be used to confirm proper equipment operation prior to and following reliability testing. The satisfactory operation test shall also be performed weekly to verify equipment operation. The satisfactory operation check of 4.5.a shall be used to monitor proper operation of the equipment daily.

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4.4.6.2.3 Reliability failure actions, Group F. In the event of failure(s) during reliability testing, failure action shall be taken as required in MIL-STD-781, Para. 5.5.

4.4.6.2.4 Corrective action, Group F. In the event the reliability test reaches a reject decision corrective action shall be taken as required in MIL-STD-781, Para. 5.7.

4.5 Examination and test methods. Examination and test methods shall be in accordance with MIL-T-28800 and as specified herein (See 6.2.2.5, 6.2.2.6, 6.2.2.7, 6.2.2.8, 6.2.3.3, Tables I, II, and III). The equipment shall be tested for the applicable class requirements of MIL-T-28800. The time base testing shall be in accordance with Figure 1. Equipment operation shall be verified in accordance with a and b below:

a. Satisfactory operation check. The satisfactory operation check shall confirm the proper operation of the internal time base, the input circuits of at least one channel, and the display section of the equipment.

b. Satisfactory operation test. The satisfactory operation test shall confirm the specified operation of the equipment for all parameters of the Level A performance test (See Table II).

4.5.2 Alternate power source testing. Testing for the alternate power sources shall be in accordance with the electrical power tests of MIL-T-28800 except the alternate power source shall be used.

4.6 Optional quality assurance testing. The procuring activity may substitute a Department of Defense (DoD) approved supplier's quality assurance system for any or all tests Table I.

4.7 System safety engineering inspection. A visual inspection shall be performed in the presence of a Government safety representative to determine that all requirements of 3.2 have been incorporated in the equipment. Inability to meet these requirements shall constitute a failure of the test.

5 PACKAGING

5.1 Preservation, packaging. Preparation for delivery shall be in accordance with MIL-T-28800 unless otherwise specified by the procuring activity (See 6.2.1.7).

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6 NOTES

6.1 Intended use. Equipment described by this specification is intended for use in testing equipments and systems in the world-wide natural and controlled environments in which military equipments are operated (on a continuous or intermittent basis for extended periods of time), stored or transported, including utilization in all types of moving military and commercial vehicles on land, sea, and air. The environmental classes and other requirements of this specification represent the majority of clearly discernable levels of the military environment within which an equipment will be used.

6.2 Ordering data. The procurement documents should specify:

6.2.1 Procurement requirements.

6.2.1.1 Title, number, and date of this specification.

6.2.1.2 Power cables if operation from 230V is desired (See 3.5.3.1).

6.2.1.3 EMI option if other than 3.8.

6.2.1.4 Maintenance and calibration aids option (See 3.10.2.1).

6.2.1.5 Time base option if other than 3.11.2.1.

6.2.1.6 First Articles, when required (See 4.2.1).

6.2.1.7 Levels of packing and packaging (See 5.1).

6.2.1.8 Warranty.

6.2.2 Contract data requirements.

6.2.2.1 Nomenclature assignment (See 3.7 and 3.10.2.1)

6.2.2.2 National stock number (See 3.7).

6.2.2.3 First Article test plans, when required (See 4.3).

6.2.2.4 First Article test reports, when required (See 4.3).

6.2.2.5 Test equipment approval requests (See 4.5)

6.2.2.6 Pretest performance records (See 4.5).

6.2.2.7 Maintainability test records (See 4.5).

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6.2.2.8 Reliability test records (See 4.5).

6.2.2.9 Quality assurance test plans (See 4.4).

6.2.2.10 Quality assurance test reports (See 4.4).

6.2.3 Acquisition management systems.

6.2.3.1 Reliability (See 3.9).

6.2.3.2 Maintainability (See 3.10).

6.2.3.3 Calibration system (See 4.5).

6.3 Definitions.

6.3.1 Fractional frequency error. Fractional frequency error is defined as the change (produced by any cause) in frequency divided by the average frequency (before the cause). For example in determining the line voltage stability, if the standard output frequency were 1×10^{7} MHz and if the standard output frequency varied by 10 Hz for any line voltage change within the specified steady-state band then the ffe for the line voltage would be 10 divided by 1×10^{7} which is equal to 1×10^{-6}.

6.3.2 Short term stability. The short term stability is defined as the square root of the Allan variance of the oscillator under test divided by the average frequency of the oscillator under test. The band width shall be 30 KHz

f_o = average frequency of oscillator under test.

$\bar{f}_k$ = k^{th} frequency sample (a frequency sample is a sample of the frequency averaged over 1 second)

M = quantity of adjacent frequency samples.
(M = 40 for this specification)

$$\text{Short term stability} = \frac{1}{f_o} \sqrt{\frac{1}{2(M-1)} \sum_{k=1}^{M-1} \left(\bar{f}_{k+1} - \bar{f}_k\right)^2}$$

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6.3.3 Sample rate. Sample rate is the time interval between the trailing edge of one gate pulse and the leading edge of the next gate pulse.

6.3.4 Front panel sample. A front panel sample is without holes, is 100mm (4 inches) by 100mm (4 inches), and is marked with the same marking process to be used on production equipment.

6.3.5 Production lot size. The production lot size shall coincide with the monthly delivery schedule quantity.

6.4 Reliability rationale. Reliability rationale submitted should provide clear and concise rationale showing how the reliability of the equipment complies with minimum requirements of the solicitation. This may include data from previous reliability tests, reliability predictions and other data available to the offeror. In the absence of such data on the equipment being offered, such data on similar equipment of equal or greater complexity produced by the offeror may be submitted. However, such data must be clearly identified as comparative data, and be accompanied with specifications and technical literature on the similar equipment.

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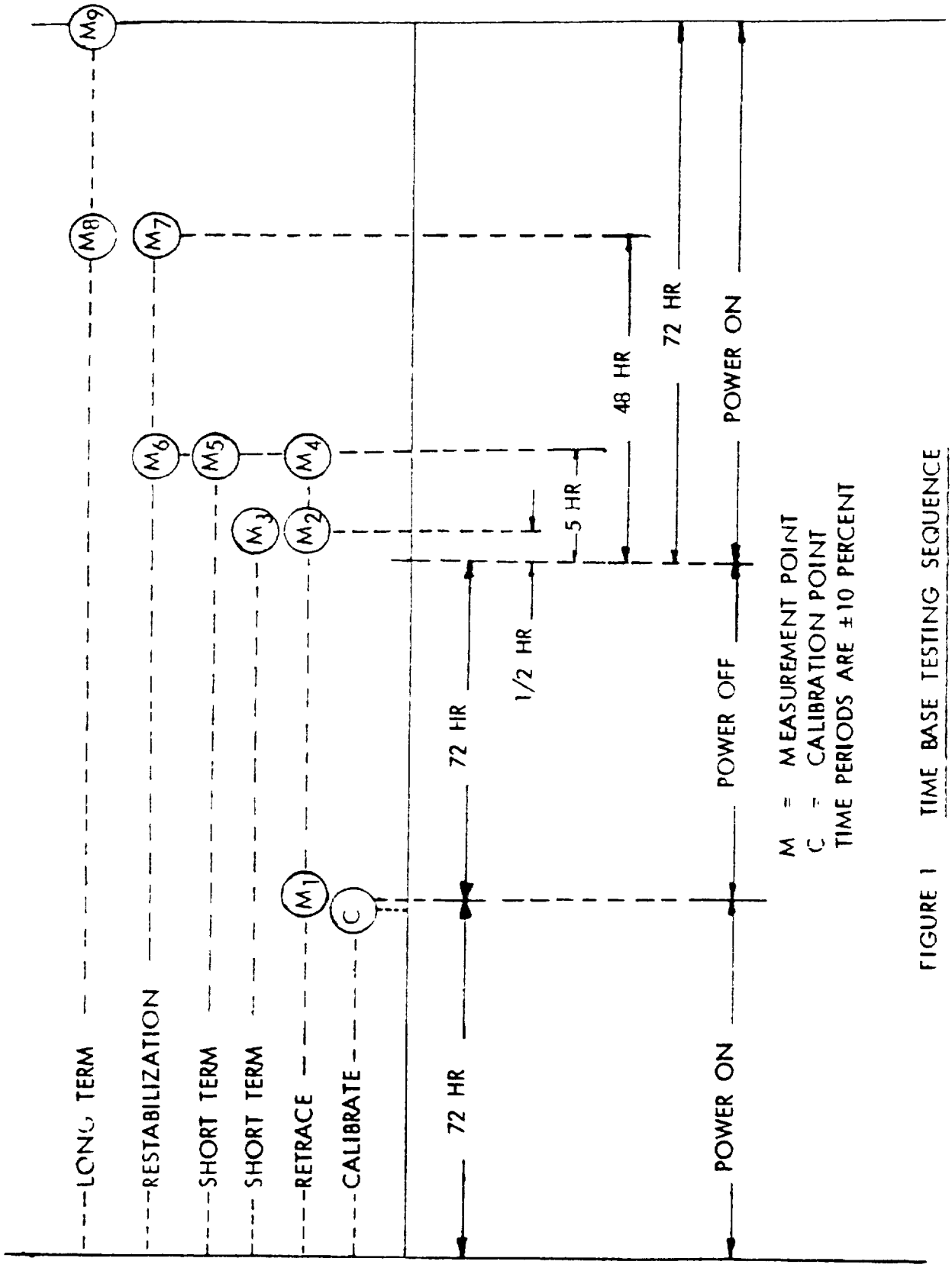


FIGURE 1 TIME BASE TESTING SEQUENCE

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